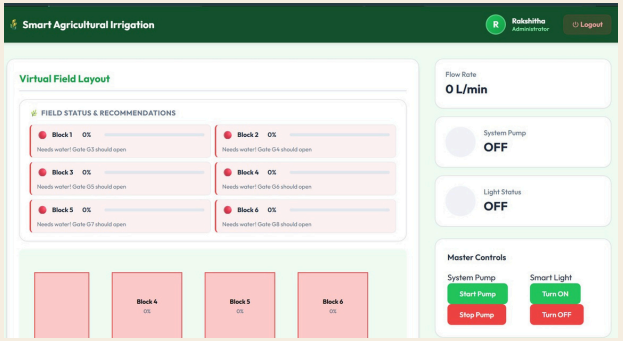
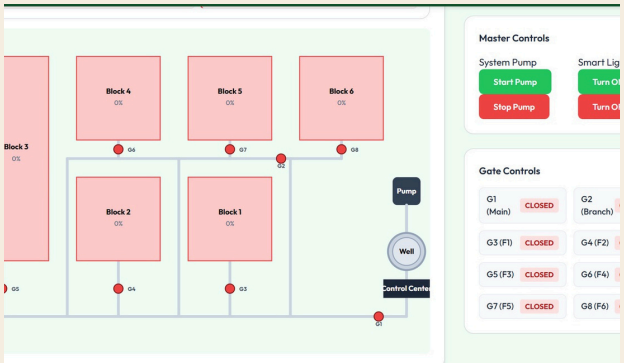
OUR- PRODUCT :

SMART FLOW





WHY TO USE

TEAM :

SMART DROPS

Team Members

MOHAMED ARSHATH M
Team leader : Hardware Designer

SANJAY K
Hardware Designer

RAKSHITHA KS
Frontend designer

SONIKA M
Backend Developer

VAISIKA P
Aiml model designer

SAI RAGAVI A C
Research and paper

Team Mentors

ANISH KARTHICK
Student Mentor

SRIRAM R
Faculty Mentor

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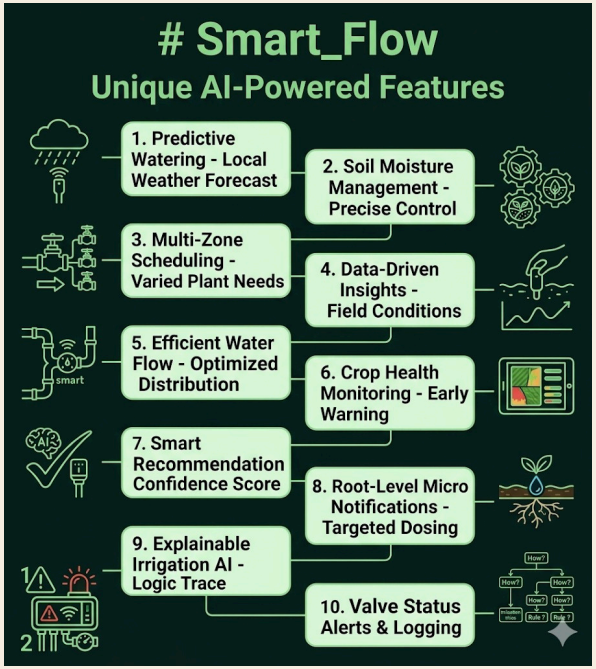
PY-EXPO

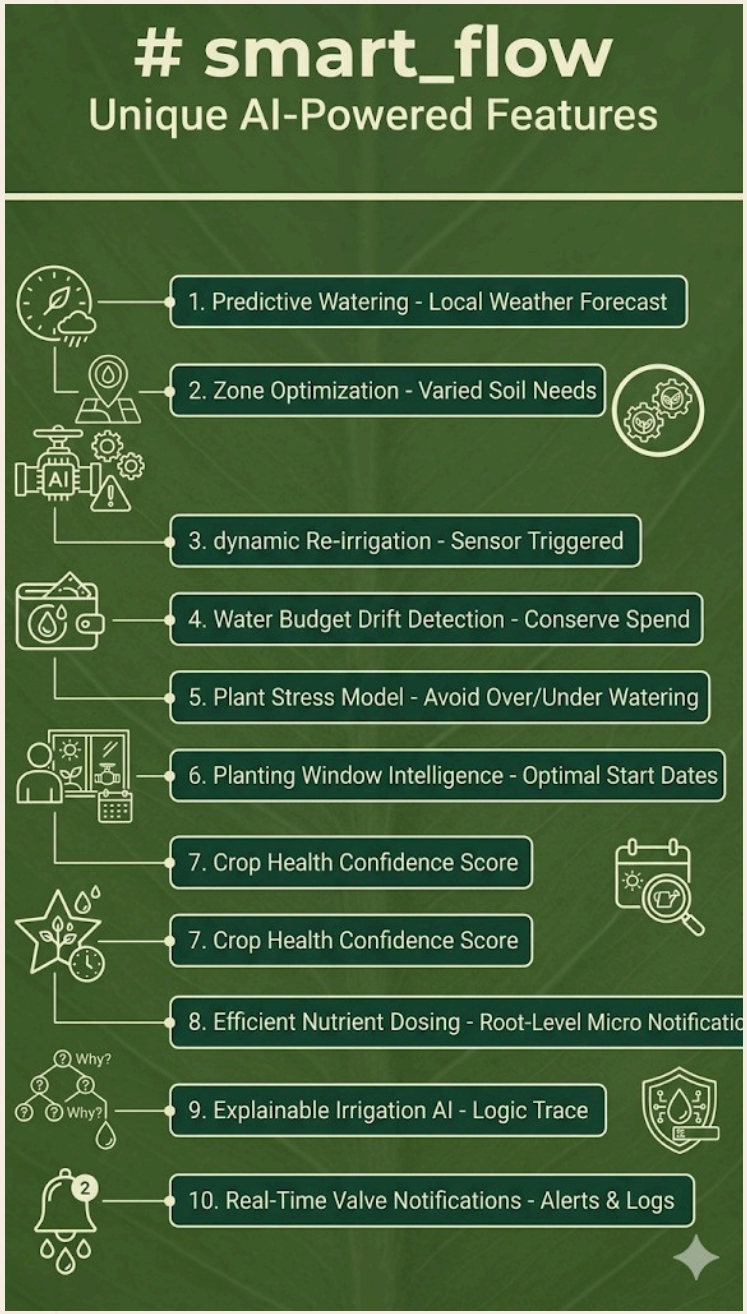
PYEXPO :

SMART DROPS



PROBLEM STATEMENT

Traditional irrigation methods leads to heavily on manual operation and fixed schedules, which often lead to overwatering or underwatering of crops. This results in inefficient water usage, increased labor effort, and reduced crop productivity. Additionally, farmers lack real-time information about soil and environmental conditions, making it difficult to make accurate irrigation decisions. There is also no proper mechanism to detect issues such as water flow failure or system faults. Therefore, there is a need for a smart and automated irrigation system that can monitor field conditions in real time, optimize water usage, reduce manual work, and provide remote monitoring and control for efficient agricultural management.

OBJECTIVE

The objective of the Smart Irrigation System is to develop an IoT-based solution that integrates both surface and drip irrigation to optimize water usage. It monitors soil moisture and environmental conditions in real time and automatically controls irrigation to prevent overwatering and underwatering. The system also enables remote monitoring, reduces manual effort, detects faults, and improves crop productivity, ensuring efficient and sustainable farming